IILM UNIVERSITY

Data Analytics Internship

Customer Segmentation & E-Commerce Analytics

Student

Asmi Dagar

Enrollment Number: CS-23411203

Program & Semester: **B.**tech CSE (DS and BD) 7th Sem

IILM University

Internship

Organization: Skillbit Technologies

Role: Data Analytics Intern

Duration: May – July 2026

Projects: Customer Segmentation & E-Commerce Funnel Analysis

About the Organization & Internship

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Organization

Skillbit Technologies

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Role

Data Analytics Intern — Intern ID: **6210**

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Duration

13th May 2026 – 12th July 2026

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Projects Completed

Customer Segmentation using K-Means Clustering

E-Commerce Funnel Analysis

Internship at a Glance

2

Projects

Completed end-to-end

60

Days

Full internship duration

Internship Objectives

The internship was structured around two end-to-end analytics projects, each designed to apply academic knowledge to real business problems using industry-standard tools and workflows.

1

Data Loading & EDA

Ingest raw datasets, perform exploratory data analysis, and identify patterns

2

Modeling & Analysis

Apply machine learning (K-Means) and funnel analytics to extract structure

3

Visualization

Build cluster plots, PCA projections, funnel charts, and campaign dashboards

4

Business Insights

Translate findings into actionable recommendations for marketing and growth

Project 1

Customer Segmentation using K-Means Clustering — identify distinct customer groups from mall transaction data

Project 2

E-Commerce Funnel Analysis — diagnose conversion drop-offs and evaluate campaign ROI across a 90-day period

Project 1: Customer Segmentation (K-Means)

Dataset & Methodology

- **Dataset:** 200 mall customers
- **Features:** Annual Income & Spending Score
- **Method:** K-Means Clustering
- **Optimal K:** 5 clusters (Elbow Method)
- **Silhouette Score:** 0.554 (strong separation)

Tools Used

- Python, pandas, scikit-learn
- StandardScaler, PCA

5 Customer Segments Identified

Wealthy Conservatives

High income, low spending

Impulsive Spenders

Low income, high spending

Average Customers

Mid income, mid spending — 40.5% of base

Prime Targets

High income, high spending — 11% of base

Careful Savers

Low income, low spending

Project 1: Insights & Business Value

Key Analytical Findings

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Income ≠ Spending

Near-zero correlation between annual income and spending score — income alone cannot predict purchase behavior

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Prime Targets (11%)

Most valuable segment — high income AND high spending. Recommended for VIP loyalty programs and premium offers

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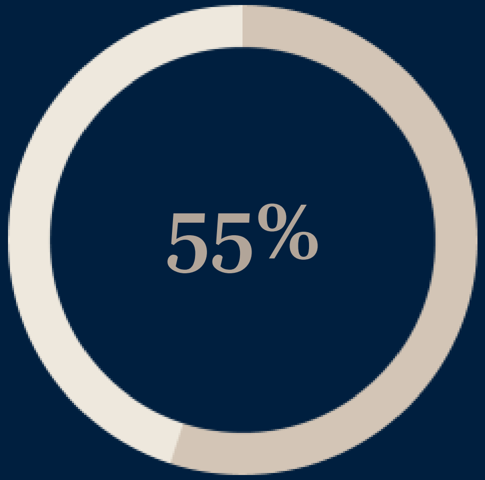
Average Customers (40.5%)

Largest segment — backbone of revenue. Best suited for seasonal campaigns and volume-based promotions

Deliverables Produced

- Cluster scatter visualizations (Income vs. Spending)
- Elbow curve (optimal K selection)
- PCA 2D projection plot
- Segment-labeled CSV exports

Silhouette Score



0.554

Strong cluster separation confirmed

Project 2: E-Commerce Funnel & Conversion Analysis

Dataset Overview

- **Sessions:** 5,000 simulated
- **Customers:** 2,257 unique
- **Period:** 90-day window
- **Funnel:** Page View → Purchase

Tools Used

- pandas, numpy
- matplotlib, seaborn

Core Metrics

3.68%

Conversion Rate

Overall funnel conversion

$13.8K

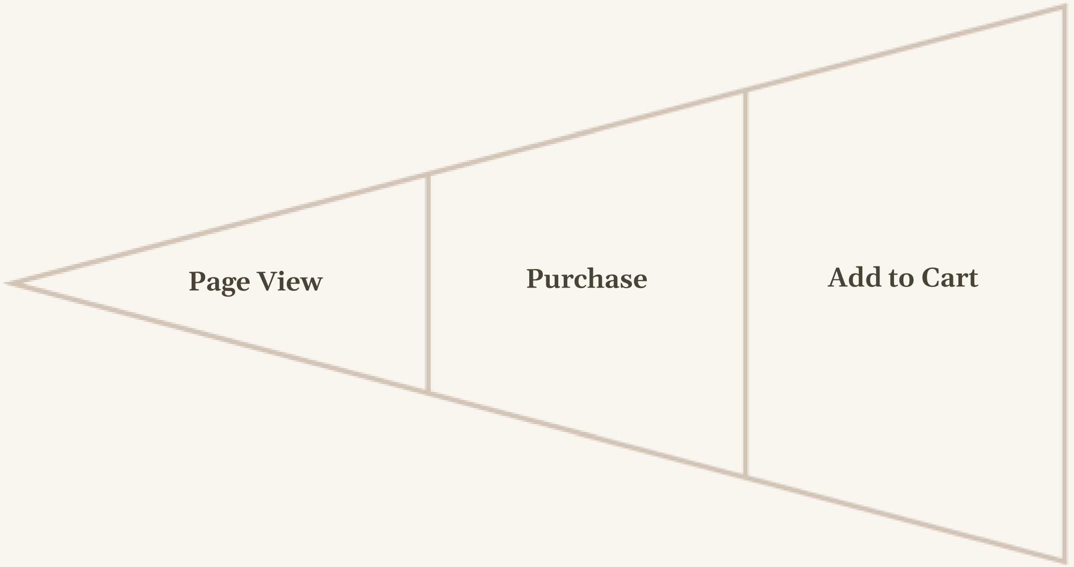
Total Revenue

$13,847.92 over 90 days

$75.26

Avg Order Value

Per completed purchase



Project 2: Key Findings & Recommendations

Critical Findings

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Biggest Funnel Leak

50% drop-off at homepage — visitors are not progressing past the landing page

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Device Performance Gap

Mobile converts **38% worse** than Desktop — significant UX friction on mobile

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Campaign ROI Disparity

Retargeting_Q1 = 291% ROI (best performer); Influencer_Push is losing money

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Customer Type Advantage

Returning customers convert **46% better** than new visitors

Strategic Recommendations

1 Fix Mobile UX

Redesign mobile checkout flow to close the 38% conversion gap vs. desktop

2 Boost Retargeting Budget

Scale Retargeting_Q1 spend — 291% ROI justifies significant budget increase

3 Pause Influencer Spend

Influencer_Push campaign is cash-negative — reallocate to retargeting

4 Launch Loyalty Program

Returning customers convert 46% better — formalize retention with rewards

Skills Gained & Conclusion

Technical Skills Developed

Machine Learning

K-Means clustering, StandardScaler, PCA dimensionality reduction

Analytics

Funnel analysis, ROI & CPA calculation, conversion rate optimization

Reporting

Dashboarding, cluster visualization, business reporting & recommendations

Internship Summary

- **Duration:** 13 May – 12 July 2026
- **Organization:** Skillbit Technologies (Intern ID: 6210)
- **Status:** Successfully completed — certificate awarded

Thank You

This internship provided hands-on experience applying data analytics and machine learning to real business challenges — from customer segmentation to conversion optimization.

Asmi Dagar

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Skillbit Technologies